

CS 6301: Wireless Vehicle V2X Final Report

Luke Castro
ldc140030@utdallas.edu
The University of Texas at Dallas
Richardson, Texas

Brandon Marzik
bnm130330@utdallas.edu
The University of Texas at Dallas
Richardson, Texas

Ricardo Alanis
Ricardo.Alanis@utdallas.edu
The University of Texas at Dallas
Richardson, Texas

Bradley Olbrey
bro160030@utdallas.edu
The University of Texas at Dallas
Richardson, Texas

Abstract

Wireless V2X networks and VANETs (Vehicular Ad Hoc Networks) are used to provide coordination and real-time responses for groups of autonomous vehicles. These decentralized networks present a unique attack space that has physical implications for vehicles and their passengers. This project will identify key points of interest within V2X networks and research the feasibility of attacks against them.

1 Introduction

The main focus of our approach to this project is to research the feasibility of various attacks against V2X networks. Our initial interpretation of V2X networks is that of an environment where endpoints may join and leave the network at any point. Given this continuous connection of new devices, we would like to research the possibility and feasibility of attacks on continuously changing networks like V2V/V2X networks.

2 Plan

We will begin by installing and configuring the necessary software (VeINS) to perform V2X network and road simulations. As we familiarize ourselves with this framework we will also use this time to better understand the standard protocols used in V2X networks and the underlying network infrastructure that allows V2X communication. With a better understanding of the protocols we will then begin researching and testing the feasibility of various attacks against V2X networks. Potential attack vectors include packet tampering, Man-In-The-Middle attacks, the manipulation of authentication in V2X networks, and various forms of denial of service. As we study the feasibility of these attacks we hope to successfully implement as many examples of attacks as we can.

3 Timeline

- 02/24: Set up VeINS system and explore the system, gain familiarity with the network
- 03/20: Build simulation in VeINS
- 04/01: Team familiar with framework and cloning process, begin vulnerability research
- 04/03: Midterm report
- 04/24: Test and validate proposed attack
- 05/01: Final Report

4 Midterm Review

4.1 Familiarization with VeINS Framework

We took the first month of our project to spend time with the simulation software we are using and get familiar with its operation and how to leverage it effectively. The VeINS framework itself is built on top of two other frameworks in order to function properly, OMNeT++ and Sumo. We had to take time and individually learn both of these smaller tools to understand how the VeINS framework works in total.

Sumo is the traffic simulation framework that VeINS uses to simulate vehicles driving around and interacting with the infrastructure around it. Sumo is set up through various configuration and XML files to define the parameters of the simulation. OMNeT++ is the simulation software that takes the VeINS framework and exchanges data between vehicles in the simulation and the sumo traffic network. With a lot of moving parts it took us quite a bit of time to get a good grasp on how everything works. We still do have issues in some areas when working with these technologies due to its complexity; while the documentation is comprehensive on individual components, examples on how to actually use them in a tied-together simulation are basic and scarce.

4.2 802.11p and wireless beaconing

VeINS uses a model of 802.11p that has been refined for use in vehicular networks. This constitutes the basis for the DRSC/WAVE standard which denotes Wireless Access in Vehicular Networks. We have implemented a working simulation to test the functionality of wireless beaconing in vehicular networks. Vehicles periodically emit beacon messages with requested application data which is transmitted to all vehicles in the network. Cars with relevant information may respond to these messages. This is just one example

of many types of communication that can occur in a vehicular network. We chose to focus on beaconing because it is a broadcast that can be read from both vehicles and road side units.

4.3 Set up working standard simulation

We have currently stood up a working simulation under normal conditions inside of the OMNeT++ simulator. We have our own road network defined and all of the standard operations and routes. Our plan is to use this current simulation as the environment of our attack.

We did have many issues with this part due to the current lack of documentation on how to use VeINS, leaving us unable to accurately diagnose error messages. When trying to make the simulation, we ran into lots of errors, including failure to load and seg-faulting. We eventually arrived at workarounds, but the occasional issue still comes up.

4.4 Proposed methods of attacks

For a DOS attack the goal is to interrupt the service of your intended target, typically by overloading the target or destroying what it is trying to send. In the case of wireless vehicles they ping each other when they are driving down the road and they also ping base stations called road side units (RSU).

We wanted to develop a threat scope that would allow for minimal interference with current implementations and be the most feasible to set up in an actual environment. For the threat scope of the attack we are creating, we define that the adversary has no access to anything currently on the network. That is, the adversary is not able to hack into or break into any vehicle or RSU that is already deployed. We define that the adversary has the capability to introduce new equipment into the network that can add more data/messages into the current network.

We currently have two methods of attack that we are testing that both start from the same assumption. We assume that the adversary is able to plant a device on the side of the road that acts like a RSU in a malicious manner. The first attack would DoS the vehicles in the simulation by overloading the vehicles with messages, causing the vehicles to not be able to accept messages. The second approach is to turn the gain up on the radios to generate noise high enough so that it distorts legitimate messages in the network.

5 Results

5.1 RSU Implementation

We implemented our RSU and configured it to reflect received messages back to the cars they came from. For our first DOS attempt, we set this message reflection in a for loop, intending for the RSU to send an avalanche of messages back to the car. Further experimentation with this approach showed that the RSU was unable to send a message to a vehicle until the previous sent message had been processed.

Investigating this, we determined that this was a property inherent to the operation of OMNeT++ itself. Initially, we expected each object and the overarching simulation to run on multiple threads, which would better simulate the unpredictability of real-world network communications. However, OMNeT++ is a single-threaded

application. This makes it deterministic in the sense that, when one object sends a message, the rest of the simulation is essentially on hold until the message is processed by the receiving object. Our original attack was meant to operate by flooding the vehicles with messages. The intent was to prevent the vehicle from participating in the rest of the network, blocking it from receiving other potentially important messages while in motion. However, the simulation was unable to support such a scenario due to its inherently deterministic design. For these same reasons, our second approach of turning up gain on radios would have no effect, as there would never be more than two messages in the simulation at the same time.

5.2 Issues with OMNeT++

When working with the simulator we found several issues that impeded our security testing. One large issue being the discrete and mostly deterministic nature of OMNeT++. Throughout our project we found odd behaviour in how OMNeT runs its simulation in terms of queuing of events. During the simulation it does not queue events at the time it comes in, but instead has certain events supersede others. For example if vehicles are queued to move soon in the simulation, but a beacon comes out then it processes that event before already queued events. For a DOS attack a realistic simulation is needed and not one that reorders events that happen.

The second issue was that the simulation conducts itself in a mostly deterministic fashion. Everything in the simulation is planned out and processes events so that it will not cause a procedural error. For example in the simulation if you try to send a beacon message faster than the car would be able to process it then the simulation will not process the event that sends the message. The simulation will not allow you to add randomness or allow simultaneous events to happen that could allow potential randomness.

Lastly OMNeT++ is a single threaded application. All simulations that it runs will only be ran on a single thread and does not allow any multi-threaded libraries to be loaded. If you try to send a large amount of messages to a vehicle it will have to send every single one individually and wait for it to be received and processed before it can continue the simulation. This is not ideal for real world security testing as this does not reflect what would happen in real life.

5.3 Changes needed

While there are many barriers that OMNeT++ has to prevent reliable security testing there are some changes that could be made to help in future projects. First thing would be to multi-thread OMNeT++, while this does not sound ideal in terms of reliable simulations it is needed to test robustness of security as in the real world not everything is deterministic. Second would be to design OMNeT++ and veins to simulate concurrent events with over the air messages in transit. This will simulate what would happen in real life on the road and give reliable expectations of the security.

6 Related Work

6.1 IEEE Standard for Information technology - Specifications Amendment 6: Wireless Access in Vehicular Environments [1]

This document is the ongoing specification for the entire 802.11 network range of protocols. Inside of this specification it lists the details for the 802.11p network which is the dedicated radio band for V2X networks. This document is a useful resource as it lists all of the details about every 802.11 network which the V2X protocol should follow as well. Inside as well it has some of the initial details about the 802.11p network. While the details are small now it is what is properly defined at the moment for V2X communication.

6.2 Man-In-The-Middle Attacks in Vehicular Ad-Hoc Networks: Evaluating the Impact of Attackers Strategies [2]

This paper examines the possibility of Man-In-The-Middle (MITM) attacks against Vehicular Ad-Hoc Networks (VANET). Due to the sensitive nature of VANETs a secure environment is absolutely necessary for the exchange of messages within a network. Any critical information has the potential to be spoofed or intercepted, causing faulty information to be spread within a vehicular network. With a Man-In-The-Middle attack, a malicious actor can potentially intercept or tamper with messages exchanged between nodes in a VANET. As a metric for measuring a potential attack, the authors use messages delayed, messages dropped, and messages tampered. These allow an accurate gauge of the amount of tampering that has occurred within a VANET. We may adopt similar metrics as we evaluate potential attacks against V2X networks.

When an event occurs in a VANET, a packet typically contains important information regarding the event such as event data, event location, and the time of the event. The authors of the paper assumed this information was both readable and editable by any vehicle on the network as packets are forwarded from malicious vehicles to legitimate vehicles. The effects of these attacks were evaluated with both random attackers distributed randomly across the network and with attackers positioned in a fleet-like structure, launching a collaborative attack against other vehicles or the network as a whole. We plan to launch similar attacks with a focus on adhering to the existing protocols for V2X communication and analyzing their authentication and encryption schemes.

6.3 Authentication-Based Vehicle-to-Vehicle Secure Communication for VANETs [3]

This paper points out the obvious security vulnerabilities associated with a vehicular ad hoc network (VANET) and proposes a lightweight authentication scheme (LIAU) to mitigate the issue. The scheme proposed uses a two-layer model (Regional Authorities and registered vehicles) and a hash function to generate authentication parameters. The paper further claims that the proposed solution can resist impersonation, modification, and replay

attacks, and has sufficiently low operation time and communication cost. This paper will be useful for our analysis of the existing protocol and its current vulnerabilities.

6.4 Bidirectionally Coupled Network and Road Traffic Simulation for Improved IVC Analysis [4]

This paper defines and introduces a framework for the simulation of Inter-Vehicle Communication (IVC) protocols called Vehicles in Network Simulation (VeINS) which we will utilize to test various attacks on V2X communications. It allows coupled network and road traffic simulation, utilizing OMNeT++, a network simulator to model VANET nodes and SUMO, a road simulation package to model and simulate microscopic road traffic. Both modules execute and communicate simultaneously to create a live simulation of both network communication and road traffic patterns. This bidirectional simulation advances time in discrete steps and guarantees synchronous execution at regular intervals. We will be using VeINS for all testing and evaluation of potential attacks against V2X networks.

6.5 V2V Wireless Communication Protocol for Rear-End Collision Avoidance on Highways [5]

This paper describes a protocol used for rear end collision avoidance for vehicles. It outlines the process by which a message warning of a potential hazard can be propagated from the lead vehicle in a line to those behind, triggering evasive maneuvers. This paper will tell us what kinds of factors we can attempt to manipulate when attempting to send data to the existing Veins system to force a desired outcome.

7 Conclusion

With the rise of autonomous vehicles and smart devices the security associated with it is very high priority. Yet with the software used to test these new devices it has been designed to test functionality and not the practicality of it. With the way OMNeT++ is designed it makes it very difficult to test best security practices due to it not truly simulating real life application. While it seems that OMNeT++ is not usable for security testing it could be with some major changes to the system.

While we did not get any clear results from our experiment, we feel that with our approach and the changes listed that we could test and possibly show that the 802.11p protocol is vulnerable in its current iteration.

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